

Name and Surname: _____

Student ITS No: _____

Qualification: _____ Year of Study: _____ Semester: _____

Assignment due date: _____ Date submitted: _____

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document.

Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents

Introduction

Main Body (with relevant subheadings)

Conclusion

References

Submit the assignment in PDF format on Moodle.

Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: _____

Date: _____

Question 1

(25 marks)

- 1.1. What is the purpose of testing a program, and when is it performed? (4 marks)
- 1.2. Identify and explain five common errors in programming. For each error:
 - Define it (syntax error, logical error)
 - Provide an example
 - Suggest how it can be prevented (15 marks)
- 1.3. Provide three control structures in programming and give a brief explanation. (6 marks)

Question 2

(30 marks)

You are tasked with designing a program for a Grade Analysis Tool that calculates and displays key performance metrics for a class of students using loops.

Requirements:

1. Welcome Message:

- Display a message introducing the Grade Analysis Tool and explaining its purpose.

2. Input Student Grades:

- Use a loop to input the grades for a class of students. The program should:
 - Ask the user how many students are in the class.
 - Collect grades for each student (out of 100).
 - Validate that the grade entered is between 0 and 100. If invalid, prompt the user to re-enter the grade.

3. Analyse Grades:

- Use loops to calculate the following:
 - Total and average grade for the class.
 - Number of students who passed (grade ≥ 50).
 - Number of students who failed (grade < 50).
 - The highest grade in the class.
 - Ensure efficient use of loop structures.

4. Display Results:

- Display the following in a well-organized format:
 - Total and average grade.
 - Number of students who passed and failed.
 - The highest grade in the class.
 - A motivational message for the class (e.g., "Great performance! Keep it up!" or "Focus on improving next time.")

5. File Handling Extension:

- Save the grades entered by the user into a file named gradesAnalysis.txt.
- At the end of the program, read the file and display the grades for verification.

Question 3

(35 marks)

Using the scenario below, create a pseudocode and a flowchart to represent the system described. Ensure that your solution includes appropriate error handling for all inputs.

Case Study: Richfield College Financial Aid Eligibility Checker

Background: Richfield College offers various financial aid packages to support students in need. To streamline the financial aid application process, the college is developing a system that helps prospective students determine their eligibility for specific aid programs based on their academic performance, family income, and extracurricular involvement.

Objective: The primary goal is to provide a user-friendly program that determines a student's eligibility for financial aid and guides them through their options.

Scenario: Imagine a student, Lunga, who is applying for financial aid at Richfield College. He has completed the online application and needs to input his details into the system.

Use Case:

1. Welcome Message:

- The program displays a welcoming message to Lunga, introducing the Richfield College Financial Aid Eligibility Checker.

2. Input Personal Information:

- Lunga is prompted to enter his details: full name, student ID, and grade level.

3. Enter Academic and Financial Information:

- The program asks Lunga to input his latest GPA and annual family income.
- He is then asked about his involvement in extracurricular activities (e.g., "yes" or "no").

4. Eligibility Criteria:

- If Lunga's GPA is 3.5 or higher, and his family income is below a certain threshold, a flag is set for eligibility for the "Academic Excellence Scholarship."
- If Lunga's GPA is between 3.0 and 3.49, and his family income is below the threshold, a flag is set for eligibility for a "General Needs Scholarship."
- If Lunga participates in extracurricular activities and has a GPA above 2.5, he qualifies for "Leadership and Community Engagement Scholarship."

5. Outcome Display:

- The program recommends the most suitable financial aid package(s) for Lunga. If no eligibility is found, an appropriate message is displayed.

6. Farewell Message:

- A goodbye message that thanks Lunga for using the Richfield College Financial Aid Eligibility Checker.